

Problem Set 1

ECON 2010 – Macroeconomics

Spring 2019

The entirety of this problem set (all questions in this packet) are **due at the start of class Week 3**. Because of how we collectively grade problem sets in class, it is not possible to turn in late problem sets. Please do your best to turn in the assignment on time.

Name:

How problem sets will work in this class.

Problem sets like this will work slightly differently than what I imagine you are used to from other classes. Although you will work on your problem set independently outside of class — please do the problem set *on your own* — we will spend a good deal of time in class going over the correct approaches to solving the problems and the correct answers. This will help you better see where your understanding of the material is incomplete, and how to correctly go about solving the problems.

To go over the problem set, you will work in small groups assigned during class to share and compare your answers to a particular problem, or part of a problem. By doing so, you'll see alternative strategies that your classmates may have used to solve the problems, and explain to others why you got the answer that you did. In your small group, you will together arrive at a *group* answer to the problem. If you've all come up with the same answer on your own, this may be easy. If you've come up with different answers, you'll have to discuss amongst yourselves to figure out who correctly solved the problem (or even, perhaps, come up with a *new* solution if you realize that no one solved it correctly).

Finally, each group will share their group answer with the rest of the class. When you see other answers shared, you'll have a chance to ask your classmates and your instructor any questions that you may have about the problem. Together, as a class, we'll confirm the correct answer to each problem and the correct solution strategy.

Problem set instructions.

1. **Write your answers on a separate sheet of paper.** Do not write on this packet until we go over the problem set during class. You will use this problem set sheet to correct the answers that you provide on the separate sheet when we go over the problem set in class the day that it is due.
2. If possible, please write your final answers in pen.
3. Please write your name on this packet and on the separate sheet(s) of paper with your answers to the questions.

4. When appropriate, show your work for each problem. Even if your answer is technically incorrect, you may get many points for the work you did to arrive at that answer. If in doubt, show your work.
 5. Note that not all questions have *one* right answer. Some questions might have many correct answers.
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Problem 1 — Transit Policy in Nashville (25 points)

When governments consider new policies that raise spending, they oftentimes have to consider taxes that will pay for those expenditures. In 2018, then-Nashville mayor Megan Berry proposed a \$5.4 billion transit plan that would have expanded bus service and introduced light rail to the city in an effort to address growing traffic. In order to fund the mass transit, the city planned to levy (among others), a new sales tax.

Consider the following questions about how an economist might go about analyzing the transit plan.

- (a) There are a lot of different commuters, businesses that those commuters work for, work hours, travel preferences and spending habits that might be affected by a new sales tax. This might make it seem difficult to consider how the transit plan might affect Nashville as a whole. As economists, let's only consider how the behavior of one average commuter is affected by the changed transit options and the increased sales tax. Which principle of economic thinking is this an example of? Choose the *best* of the two answers below. **(4 points) Remember: Write your answers on a separate sheet of paper. See the problem set instructions.**

(a) Rational actors

(b) Simplifying assumption

- (b) Nashville is a dynamic city – there's a lot going on in terms of new policies, changing economy, people moving to and from the city, and more. Considering all of these changes along with a new transit plan and new sales tax is challenging. As economists, then, let's only consider the changing transit options in the transit plan and the increased sales tax while assuming that no other changes are taking place at the same time. Which principle of economic thinking is this an example of? Choose the *best* of the two answers below. **(4 points)**

(a) *Ceteris Parabis*, i.e. holding other factors fixed

(b) Simplifying assumption

- (c) Consider Chase, a man who lives in Murfreesboro south of Nashville and works in downtown Nashville. Chase commutes 1.5 hours each morning and each evening, 5 days a week. He uses 2 gallons of gas for each his morning and evening commute. Gas costs \$2 per gallon. Chase also values his time. For instance, he likes playing fetch with his dog in the morning and evening. That is, Chase faces an *opportunity cost* to sitting in his car in traffic – not playing with his dog. Let's say the value of Chase's time is \$16 per hour. (You could think of this as what he *could* earn if he wasn't in traffic, or how much he values time spent with his dog.) Given all of this information, how much does it cost Chase – in *both* – explicit costs and opportunity costs to commute in traffic to his job each day. You may think that there should be other costs considered (e.g. car maintenance, insurance, parking, etc.), but only consider the costs mentioned above. If Chase works 240 days a year, how much does it cost him to commute by car? **(5 points)**
- (d) Now consider what happens if a rail line is built between Murfreesboro and downtown Nashville. (Let's assume that Chase is lucky and there is a station right next to his apartment and one right next to his work.) The journey on the train takes 1 hour each way for Chase. A ticket is \$5 each way. Chase now has an extra 30 minutes to play with his dog each morning and each evening. Recall above that he values this time at \$16 per hour. Relative to the cost of driving you found above, how much does Chase save in both opportunity cost and regular costs per day by taking the train? If Chase works 240 days per year, how much does he save relative to commuting by car per year? **(5 points)**
- (e) Chase works in Nashville, but he never shops there. He buys everything in Murfreesboro where he lives. Shateeka, however, lives and works in Nashville. Her apartment is right down the street from her office, and she walks to work, so the transit plan doesn't help her at all. Because she lives in Nashville, she buys everything in the city – groceries, movie tickets, clothes, toiletries, furniture, etc. The current sales tax is 9.5% and would increase to 9.75% next year according to the transit plan. If Shateeka spends \$10,000 per year on items subject to the sales tax, how much more tax does she pay from before to after the transit plan? **(5 points)**

- (f) Consider the following excerpt from a fictional article on the Nashville transit plan. Select **one** sentence that represents a *positive* statement about rent control, **and one** sentence that represents a *normative* statement. (2 points)

“Estimates that as many as 100 people a day are moving to Music City mean that our roads are becoming more and more congested. Roadways are filled to capacity at rush hour and commute times keep skyrocketing. The Mayor’s office should make a proposal to fix this traffic crisis. But the current transit plan should not be adopted. The sales tax used to pay for the current transit plan falls too heavily on city dwellers rather than the suburban commuters who contribute to traffic and who will benefit from the transit plan. The tax will also discourage tourism, which is vital to the city’s economy.”

Problem 2 — Production Possibilities in a Country like Denmark (16 points)

Consider a very simple economy that produces only two goods — Legos and herring. Legos are a plastic toy that you can use to build whatever you want. Herring is a kind of disgusting ocean fish found in the northern Atlantic Ocean. The following table represents the amount of Legos that can be produced for different amounts of herring produced. Alternatively, you can think of the table as representing amount of herring that can be produced for different amounts of Legos produced.

Table 1: Production of Legos and herring in a fictional economy

Quantity of Legos, millions of lbs	Quantity of herring, millions of lbs
0	80
120	0
40	70
110	10
70	60

- (a) Use the information in the table above to draw (graph/plot) a production possibilities frontier (PPF) for this simple Legos and herring economy on an x-y plane. The x – axis on your

graph should be the quantity of Legos and the y – *axis* should be the quantity of herring. Label your axes. Do your best to draw your production possibilities frontier to scale, but of course it doesn't have to be perfect. **(5 points)**

- (b) Given the production possibilities frontier, for each of the following points, is the production of Legos and herring at this point *possible*? **(3 points total, 1 point each)**

(1) The point where the quantity of Legos is 40, and the quantity of herring is 50.

(2) The point where the quantity of Legos is 90, and the quantity of herring is 90.

(3) The point where the quantity of Legos is 69, and the quantity of herring is 59.

- (c) Given the production possibilities frontier, for each of the following points, is the production of Legos and herring at this point *efficient*? Note that you may be better off using calculations for some of these rather than just looking at your graph. **(3 points total, 1 point each)**

(1) The point where the quantity of Legos is 40, and the quantity of herring is 50.

(2) The point where the quantity of Legos is 20, and the quantity of herring is 75.

(3) The point where the quantity of Legos is 120, and the quantity of herring is 0.

- (d) Assume that there is only one input into the production of potatoes and beef: labor. That is, producers in the economy decide whether to use their labor effort to fish for herring or to manufacture Legos, and nothing else is needed (no land, no natural resources, no plastic, etc.). (Notice that this is a simplifying assumption!) Draw an entirely *new* production possibilities frontier to reflect an economy in which when 100 million lbs of herring is produced, no Legos are produced; *and* labor (the one input in this economy) is **equally as productive** in making Legos as it is in fishing for herring. Label your x and y axes the same as your previous plot, and label the point where the PPF crosses the x-axis. At this crossing point, how many Legos are produced when no herring is produced? (5 points)

Problem 3 — Fixed and Marginal Costs in the Airline Industry (20 points)

The history of the market for airline flights in the United States is an interesting one that not many people know about. Before 1978, the airline industry was heavily regulated by the federal government. The government dictated to each airline company which routes it could fly and how much it could charge for a ticket on that route.

Because airlines could not compete on price for the same route, they had to find other ways to get passengers to choose their airline over another. One way that they did this was by offering lots of flights at lots of times and offering flexible tickets to passengers. Even if your ticket was for a particular flight at a particular time, the flexible ticket meant that whenever you showed up to the airport, you would be able to get on the next plane to your destination.

In order to guarantee that passengers could get on a flight whenever they showed up at the airport, airlines had to run many more flights than they had passengers. (Otherwise you might show up at the airport to find the next planes were full! You might look for flights on another airline in the future as a result.) This resulted in a lot of nearly empty airlines flying around.

After 1978, the airline industry was de-regulated. The government stopped dictating which airlines could fly which routes and stopped setting prices. This is the (relatively) free airfare market that we're familiar with today.

- (a) It cost TWA, an airline company, \$100,000 per day in fixed costs to pay airport fees, leases on its airplanes, the reservation system, and advertising. TWA had to pay these fixed costs

regardless of how many flights it operated. That is, even if it operated no flights, it couldn't change the airport fees it had paid, the cost it had paid for its airlines, etc. In addition, it costs \$10,000 to operate each of its 10 daily flights. For each flight, this \$10,000 covers pilot and crew wages, fuel, food for passengers, and service for the planes. Let's consider what would happen if TWA wanted to add another daily flight to, say, better serve customers who wanted an early morning flight option. What would the marginal cost have been to TWA of adding this 11th daily flight? **(5 points)**

- (b) Consider the same information from part (a) above. If TWA can charged \$200 per ticket on this 11th flight, would it have been sensible (cost effective) to run this flight with only 25 passengers? Explain briefly why. **(5 points)**
- (c) Again, consider the same information from part (a) above. If TWA could have again charged \$200 per ticket on its 11th flight, would it have been sensible to run this flight with only 75 passengers? Explain briefly why. **(5 points)**
- (d) Consider the same information from part (c) above. If the fixed costs that TWA had to pay doubled to \$200,000 how would this change whether it was sensible to offer the 11th flight? Explain briefly why it would or wouldn't change whether it was sensible to offer the 11th flight. **(5 points)**

Problem 4 — Specialization in a Simple Society (31 points)

Consider an early human living a few thousand years ago. This person — let's call her Sara — needed three things to survive — food, clothing and shelter. In this simple early economy, Sara lived in isolation and had to provide all three life necessities for herself. Consider food in the form of fish that Sara had to catch, clothing in the form of fabric that Sara had to weave herself (and sewed into clothes), and shelter in the form of wood that Sara cut down from trees and chopped up (and built into a dwelling).

Table 2: Sara's Production of Food, Clothing and Shelter

Fish caught per hour	Fabric woven per hour	Trees cut per hour
8	2	1

- (a) Given the information in the table above, if Sara chooses to spend an 8 hour day cutting down trees, how many trees can she cut down? During that same 8 hour day, what is the opportunity cost of cutting down those trees in terms of fabric and fish? **(3 points)**.
- (b) In Sara's production of food, clothing and shelter in the table above, we've effectively assumed that she exhibits *constant marginal returns* to her labor. That is, the first hour she spends fishing is *just as productive* as her last hour spent fishing, just as her first hour of weaving is just as productive as her last (or any other), and the same for chopping trees. If Sara's work exhibited *increasing marginal returns* to her labor, would she be able to catch more fish or less fish at the end of a day spent catching fish? **(4 points)**
- (c) Now consider that Sara doesn't live in quite so much isolation. Instead, there's another person who has to provide himself with the same three life necessities who lives on the other side of the hill from her. Marcus is a better fisherman than Sara. In an hour, he can catch 15 fish. He is also better at weaving. In an hour, Marcus can weave 3 units of cloth, but he can only

cut down half (.5) of a tree. If Marcus chooses to spend an 8 hour day cutting down and chopping trees, how many trees can he cut down, and what is his opportunity cost of doing so in terms of fish caught and cloth woven? **(3 points)**

(d) Given the opportunity costs for Sara that you found in part (a) and for Marcus that you found in part (c), answer the following. **(6 points, 2 points each)**

(1) Who has a higher opportunity cost of chopping trees as compared to catching fish?
That is, which person has to give up catching more fish to chop down a tree?

(2) Who has a higher opportunity cost of chopping trees as compared to weaving cloth?
That is, which person has to give up weaving more cloth to chop down a tree?

(2) Who has a higher opportunity cost of catching fish as compared to weaving cloth?
That is, which person has to give up weaving more cloth to catch a fish?

(e) Now imagine that a perfect perpetual summer has rolled around (a mini global warming) where there's never any rain, snow or any wind. Neither of them have any use for shelter. Instead, they only need fish and clothing to survive. One of these balmy days, Sara and Marcus meet for the first time on the top of the hill. The conversation turns to how well each can provide for themselves — how well each can fish and how well each can wave cloth. Despite their primitive early human brains, the two think that they can *together* produce more fish *and* more cloth if each specializes in one activity. In order to maximize the total fish and total fabric that they collectively produce, who should specialize in fishing? Who should specialize in tree chopping? Explain briefly why. **(5 points)**

- (f) Sara and Marcus each had to provide for themselves before meeting. Marcus spent 5 hours fishing and 5 hours weaving each day. Sara spent 8 hours fishing and 2 hours weaving each day. Without specialization, how much fish and how much cloth could Marcus and Sara produce together in a day? **(4 points)**
- (g) If Sara and Marcus specialize as you found in part (e) above, one will only fish and one will only weave. If each of them work 10 hours per day on their one task, how many fish will they catch and how much cloth will they weave? Is it more or less than what you found without specialization in part (f) above? **(4 points)**
- (h) Is the following a positive or normative set of statements? **(2 point)**
“People shouldn’t specialize because they lose the ability to take care of themselves. Even if total production is increased, people shouldn’t specialize because they lose the ability to produce their own food, or address their own medical needs, or even do simple things like make fire.”

Correcting Your Problem Set (8 points)

You’ll receive 8 points for correcting your problem set in class — that is, marking the answers that you got correct on this sheet, and writing an explanation for why some of your answers were incorrect on others. See the section on *How Homework Will Work* on the first page the problem set.